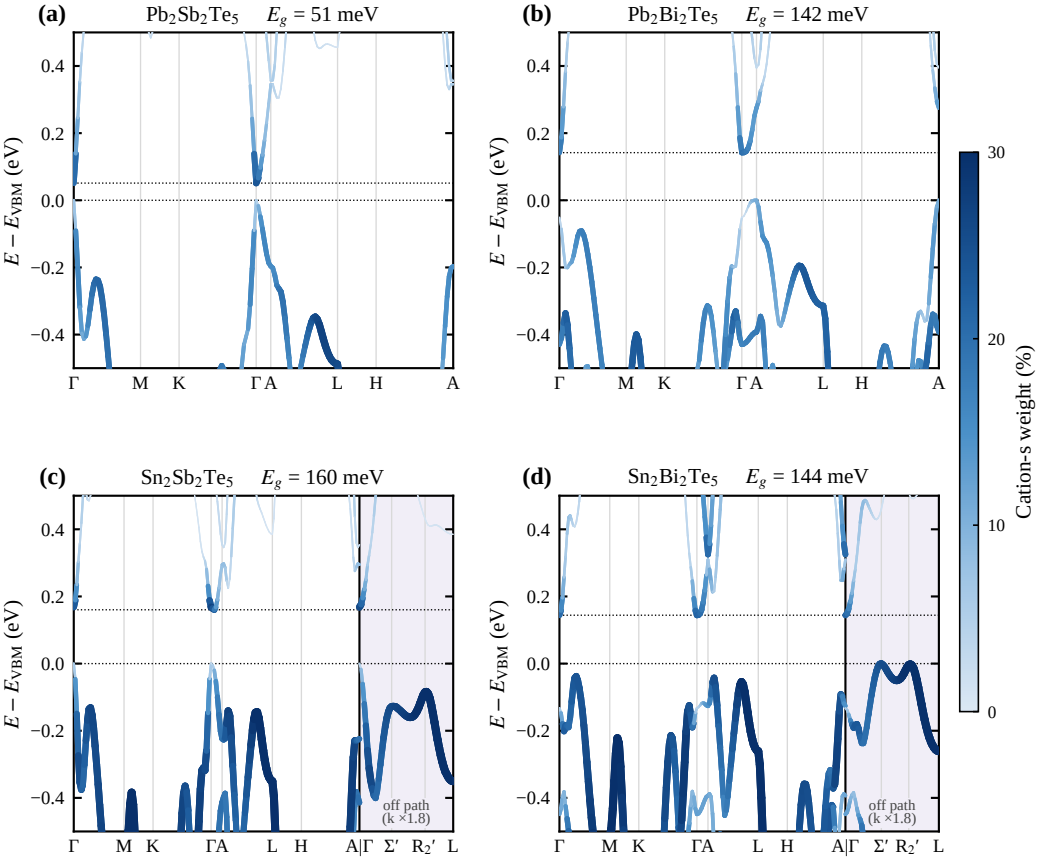

(a)
Pb2Sb2Te5
Eg = 51 meV
(b)
Pb2Bi2Te5
Eg = 142 meV
(c)
Sn2Sb2Te5
Eg = 160 meV
(d)
Sn2Bi2Te5
Eg = 144 meV
E − EVBM (eV)
0.4
0.2
0.0
−0.2
−0.4
Γ M K ΓA L H A
Γ M K ΓA L H A|Γ Σ′ R2′ L
off path
(k ×1.8)
Cation-s weight (%)
30
20
10
0